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SAP Cloud ALM: monitoring Integration Suite and SuccessFactors

Tom Rottiers, SAP Inside Track, 24/09/2022

MAKING AN IMPACT THAT MATTERS since 1845

Me in a nutshell

Tom Rottiers

Deloitte

Working in HR tech since 2000

Certified consultant in SAP R/3 4.6b ☺

Last 7 years active in architecture, integrations

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Why this presentation?

- System and integration landscapes are becoming more and more complex
- As a consequence, monitoring the integration flows has also become more complicated

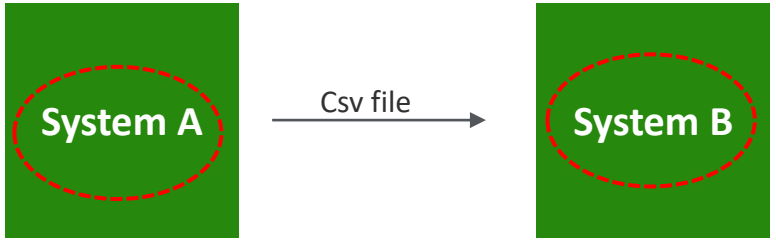
From:

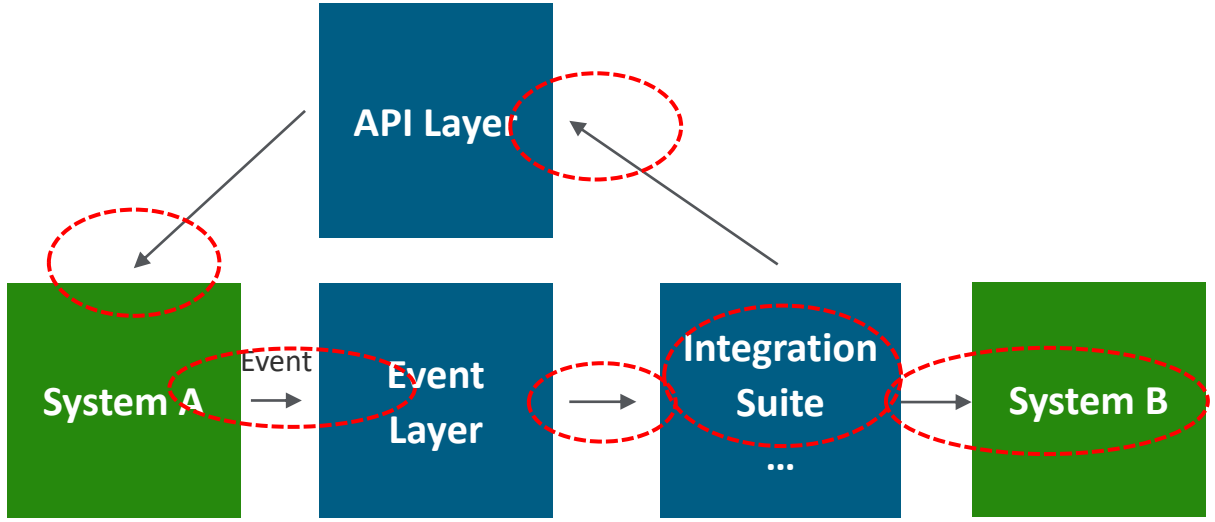
- Scheduled batch jobs
- Tightly coupled point to point integrations
- File based

To:

- (near) real time
- Loosely coupled, event-based integration
- Web service based

- Greater flexibility, but a lot more points of failure and much more difficult to have an end-to-end monitoring





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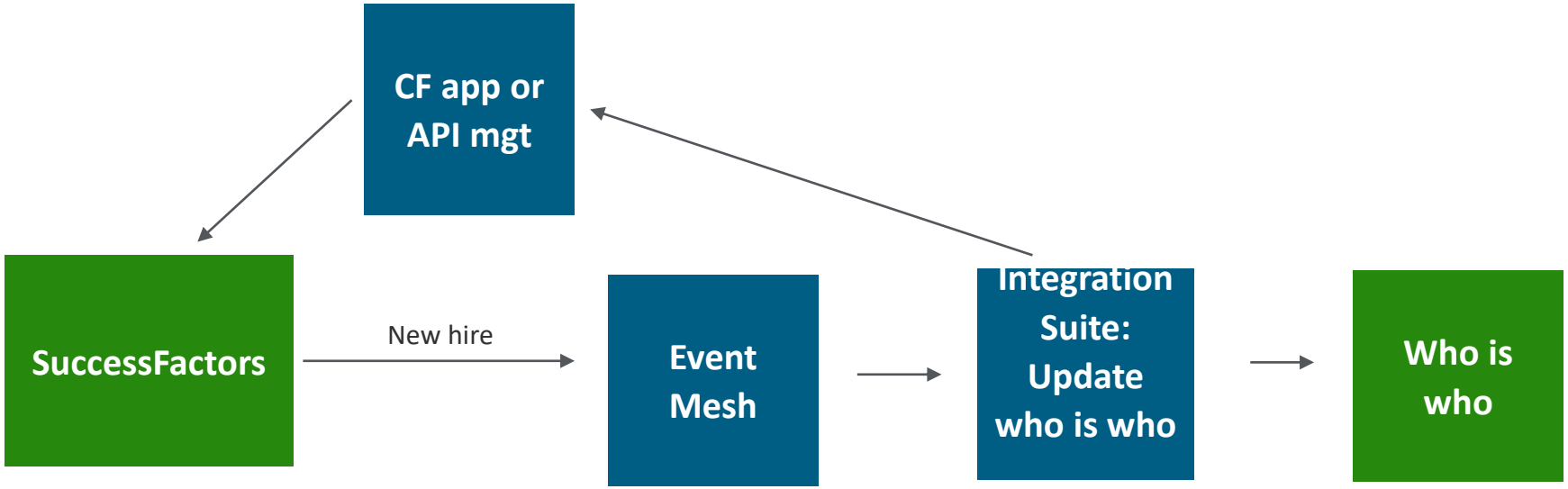
Why this presentation?

- In SuccessFactors context



- What can go wrong:
  - Event not fired or doesn't arrive in event mesh
  - Event doesn't arrive in integration suite
  - Integration suite flow crashes
  - Issues with API to fetch employee data: authentication, other exceptions
  - Errors when updating who is who

- How to monitor end-to-end?
  - In integration suite: custom error handling flows, Realcore dashboard,…
  - In CF: alert notification, dead letter queue monitoring, …
- Cloud ALM to the rescue!

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Agenda

Topic	Content
SAP Cloud ALM: introduction	• What is Cloud ALM • What does it cost + how to activate
SAP Cloud ALM for Operations	• Functionalities and configuration
API’s	

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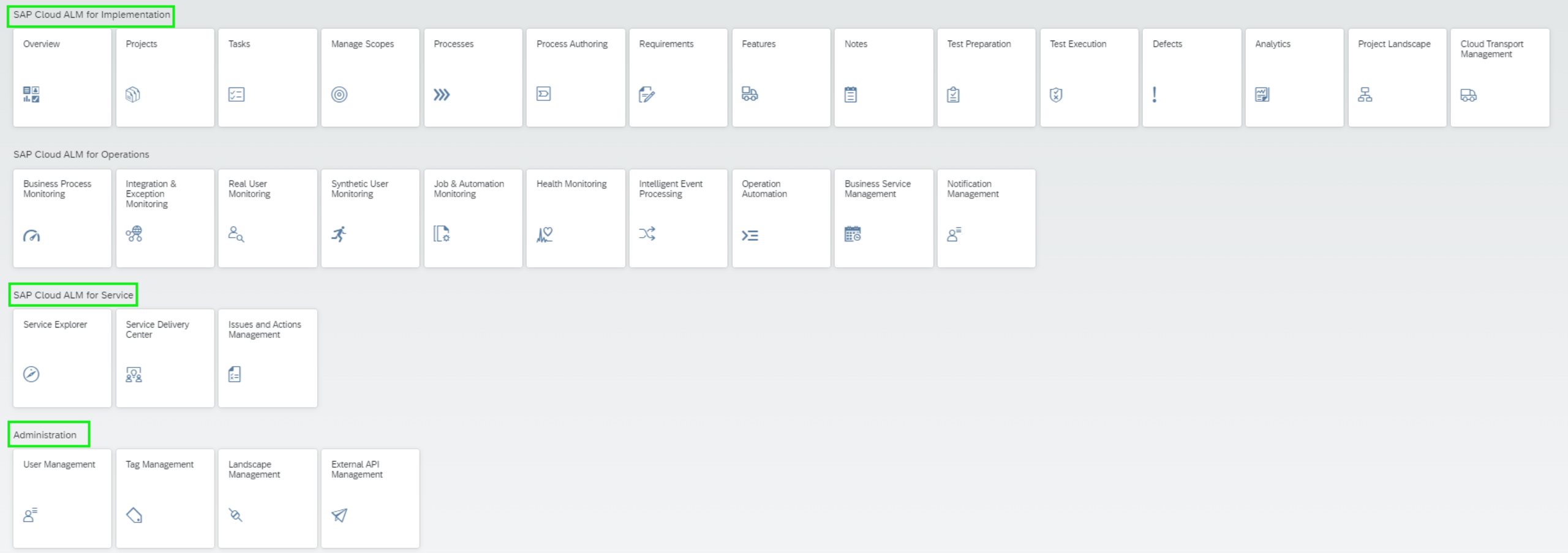
SAP Cloud ALM: Introduction

SAP Cloud ALM: introduction

What is Cloud ALM

Application Lifecycle Management offering for cloud-centric customers

Support for both implementation of projects and operations (after go live) for cloud solutions



SAP Cloud ALM: introduction

What does it cost and how to activate?

Cost

SAP Cloud ALM is included in the license for Enterprise Cloud Support.

E.g. if you have a subscription to Integration Suite, SuccessFactors that comes with Enterprise Support, you can use Cloud ALM for **FREE**

Public demo tenant available:

https://support.sap.com/en/alm/demo-systems/cloud-alm-demo-system.html

Supported solutions:
https://help.sap.com/docs/CloudALM/08879d094f3b4de3ac67832f4a56a6de/3ebab44c25da4cc1997bad08cc00991c.html

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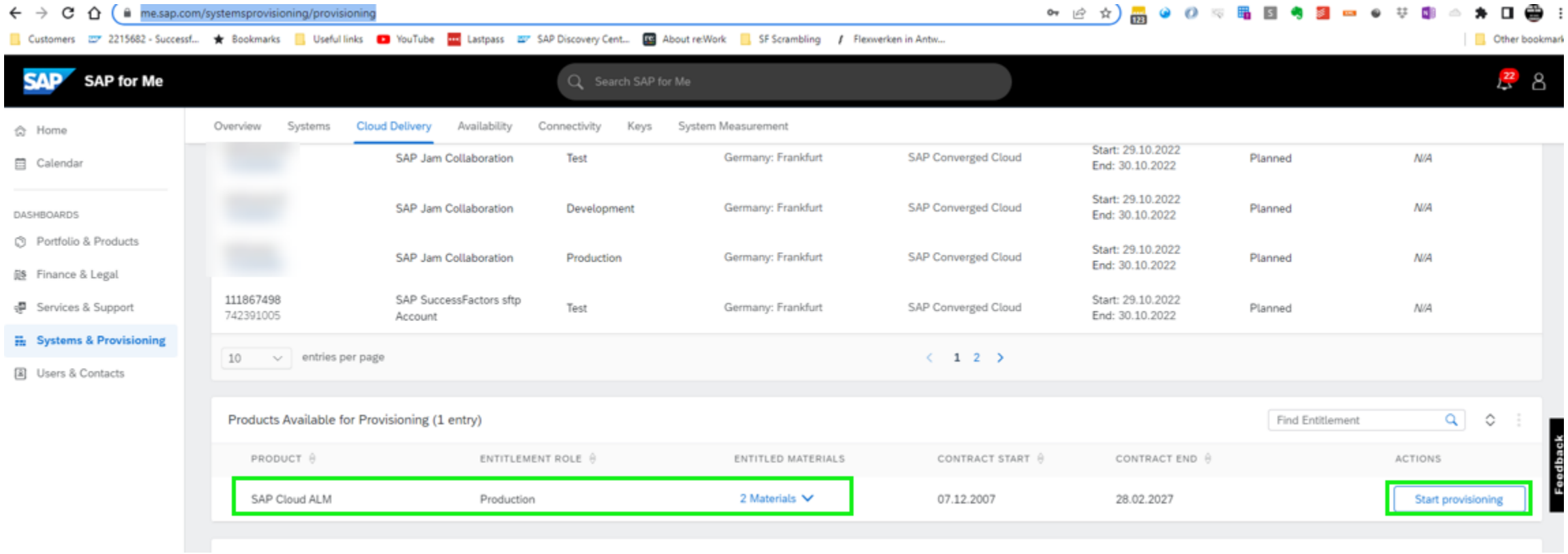
SAP Cloud ALM: introduction

What does it cost and how to activate?

Activation

Go to https://me.sap.com/systemsprovisioning/provisioning

Select tab Cloud Delivery and click on Start Provisioning



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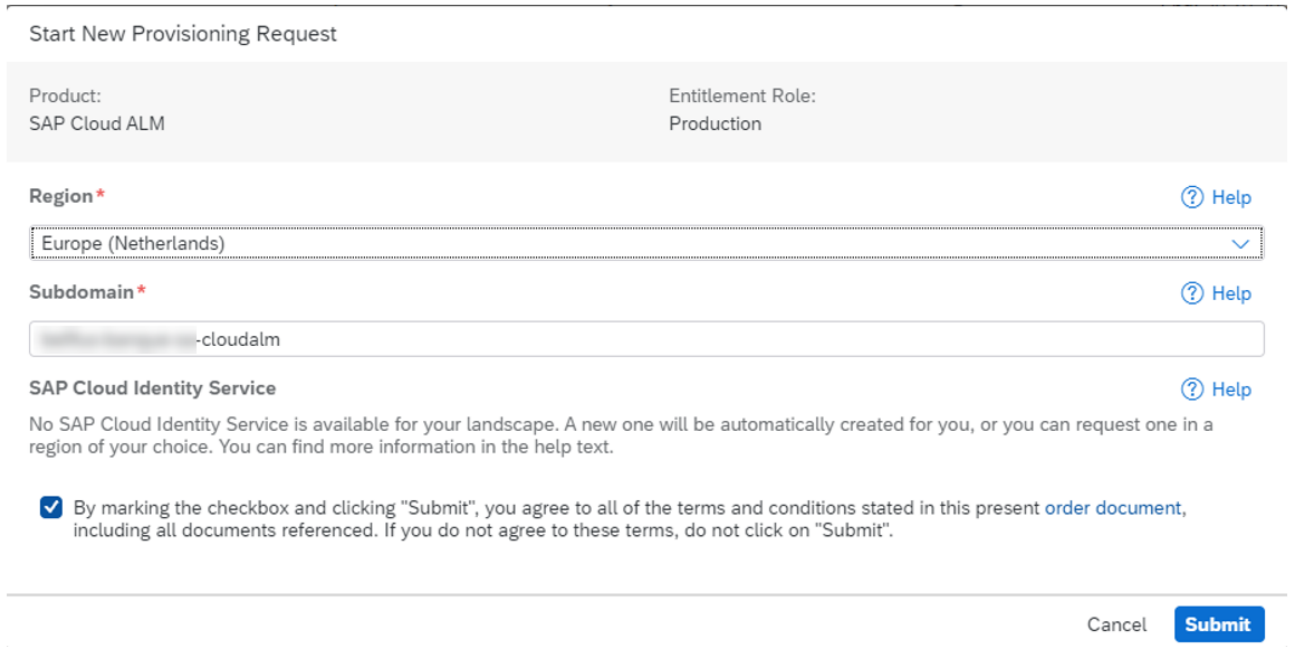
SAP Cloud ALM: introduction

What does it cost and how to activate?

Activation



SAP will perform provisioning:

Subaccount, IAS, ALM tenant

Setup guide:

https://help.sap.com/docs/CloudALM/08879d094f3b4de3ac67832f4a56a6de/cd38b1f8e2494707b6a1368b39cc8767.html

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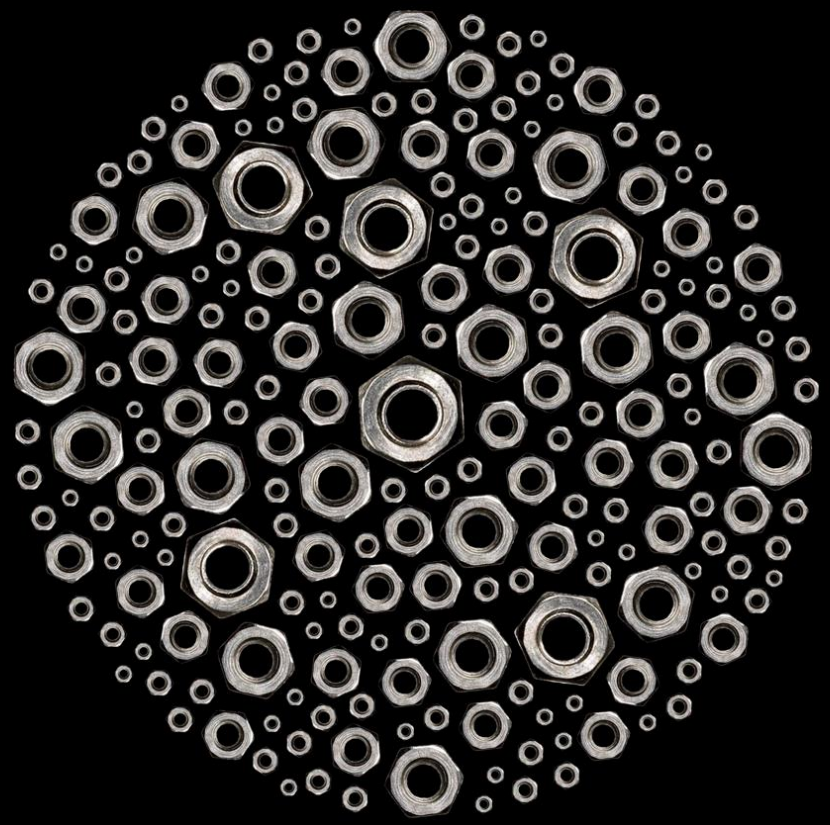
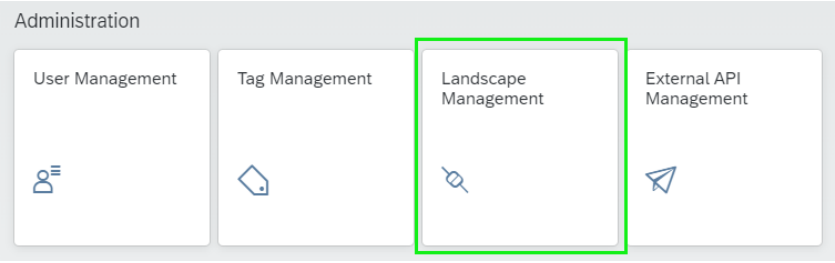


SAP Cloud ALM for operations

SAP Cloud ALM for operations

Landscape management



After activation of SAP Cloud ALM => list of available cloud systems automatically included in Landscape Management

Services

Services (32)

Live Search | Add

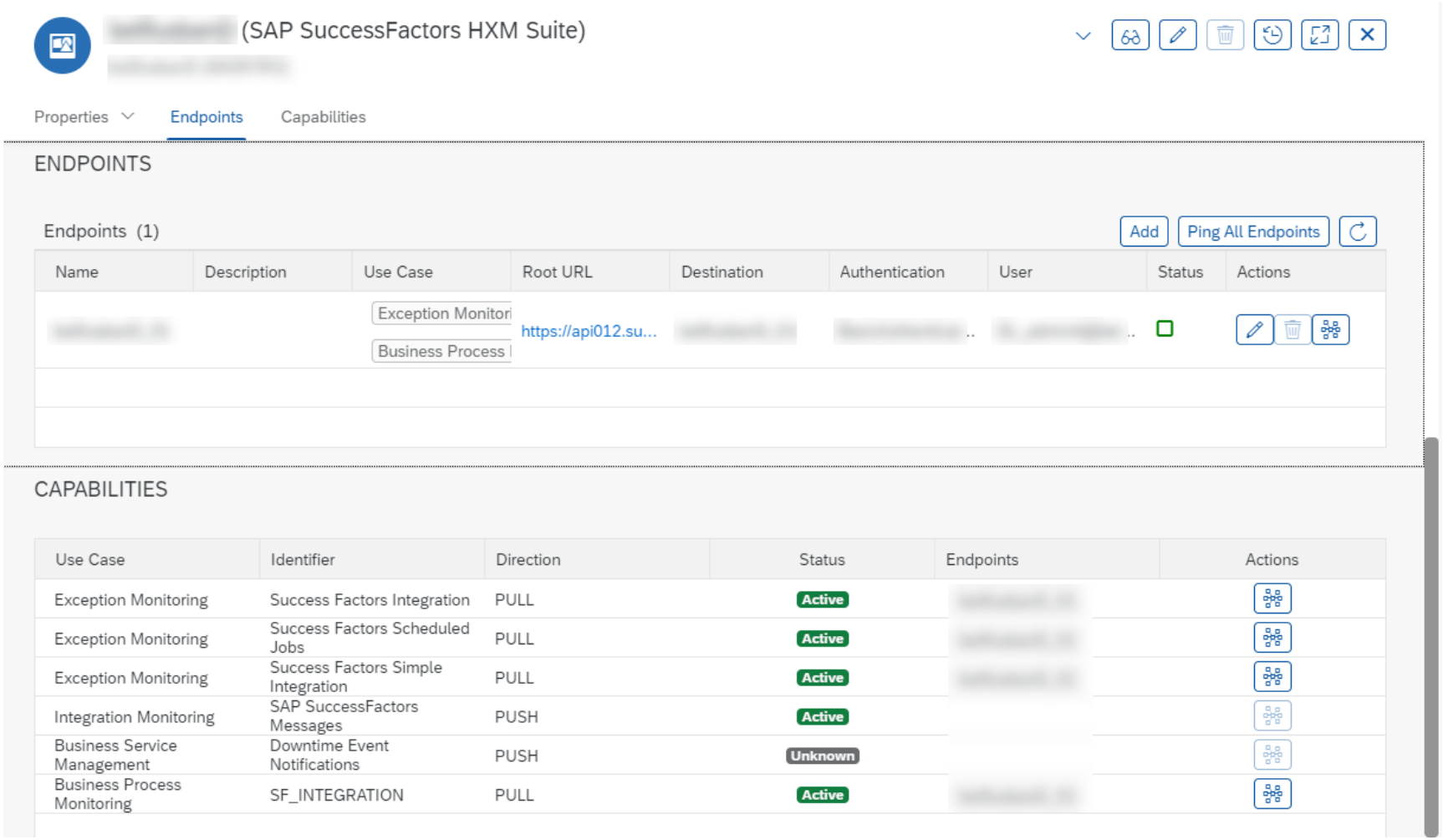
Status	Name	Service Type	Customer Name	Role	Tenant ID	Data Center Name
New		SAP BTP Neo environment		Production		Netherlands: Amsterdam
New		SAP Ariba Network		Production		USA: San Jose, CA
New		SAP Ariba Network		Test		USA: San Jose, CA
New		SAP BTP Neo environment		Production		Netherlands: Amsterdam
New		SAP SuccessFactors Recruiting Posting		Production		Germany: Walldorf
New		SAP SuccessFactors HXM Suite		Production		Germany: St. Leon-Rot
New		SAP Jam Collaboration		Production		Germany: St. Leon-Rot
New		SAP SuccessFactors Recruiting Marketing		Production		Germany: St. Leon-Rot
Active		SAP SuccessFactors HXM Suite		Development		Germany: St. Leon-Rot
New		SAP Jam Collaboration		Development		Germany: St. Leon-Rot
New		SAP SuccessFactors Recruiting Marketing		Test		Germany: St. Leon-Rot
New		SAP SuccessFactors Recruiting		Development		Germany: Walldorf

SAP Cloud ALM for operations

Landscape management

For each type of system => different services are available

- Business Process Monitoring
- Exception Monitoring
- Health Monitoring
- Real User Monitoring



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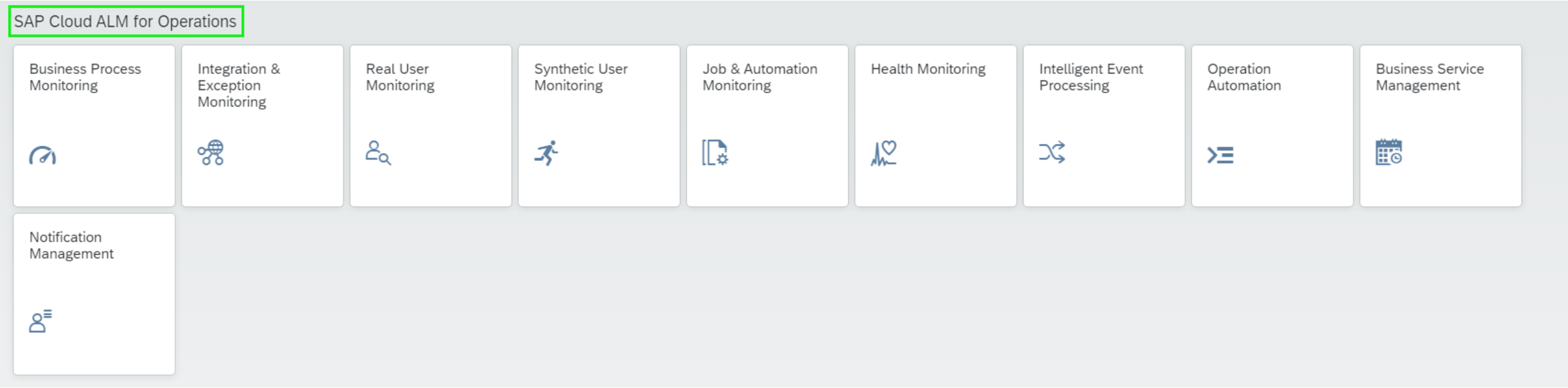
SAP Cloud ALM for operations

Introduction

- **Features, functions and processes** for operating your cloud centric solution landscapes.
- Having a **central entry point** to operate your **complete SAP landscape** running on different technologies.



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SAP Cloud ALM for operations

Introduction

Business Process Monitoring

Synthetic User Monitoring

Operation Automation

Integration & Exception Monitoring

Job & Automation Monitoring

Notification Management

Real User Monitoring

Intelligent Event Processing

Health Monitoring

Business Service Management

next

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SAP Cloud ALM for operations

Business Process Monitoring

Business Process Monitoring

Overview / Recruit to Retire

Recruit to Retire 9/17/22, 12:00 AM - 9/24/22, 6:33 AM

Country/Region: | Internal Status: | Job Requisition Progress (Candidate Label - Sta... | Job Code: | Legal Entity Code: | Go | Adapt Filters

Attract to Hire				Develop Workforce				Manage Workforce				Retire Workforce			
0 of 0	0 of 0	3 of 16	3	0 of 0	0 of 0	0 of 0	3	0 of 0	0 of 0	0 of 0	6	0 of 0	0 of 0	0 of 0	2

Job Requisitions Created

5

Job Requisitions with Applications

60

SAP Cloud ALM for operations

Integration and exception monitoring

Integration & Exception Monitoring

Overview | Favorites

Favorites

SAP Integration Suite (Cloud Integration)

Messages 359 2 84K

Exceptions 0

Quality Alerts 358

SAP SuccessFactors HXM Suite

Messages 0

Exceptions 117

Quality Alerts 0

SAP Integration Suite (Cloud Integration)

Messages 2

Exceptions 0

Quality Alerts 0

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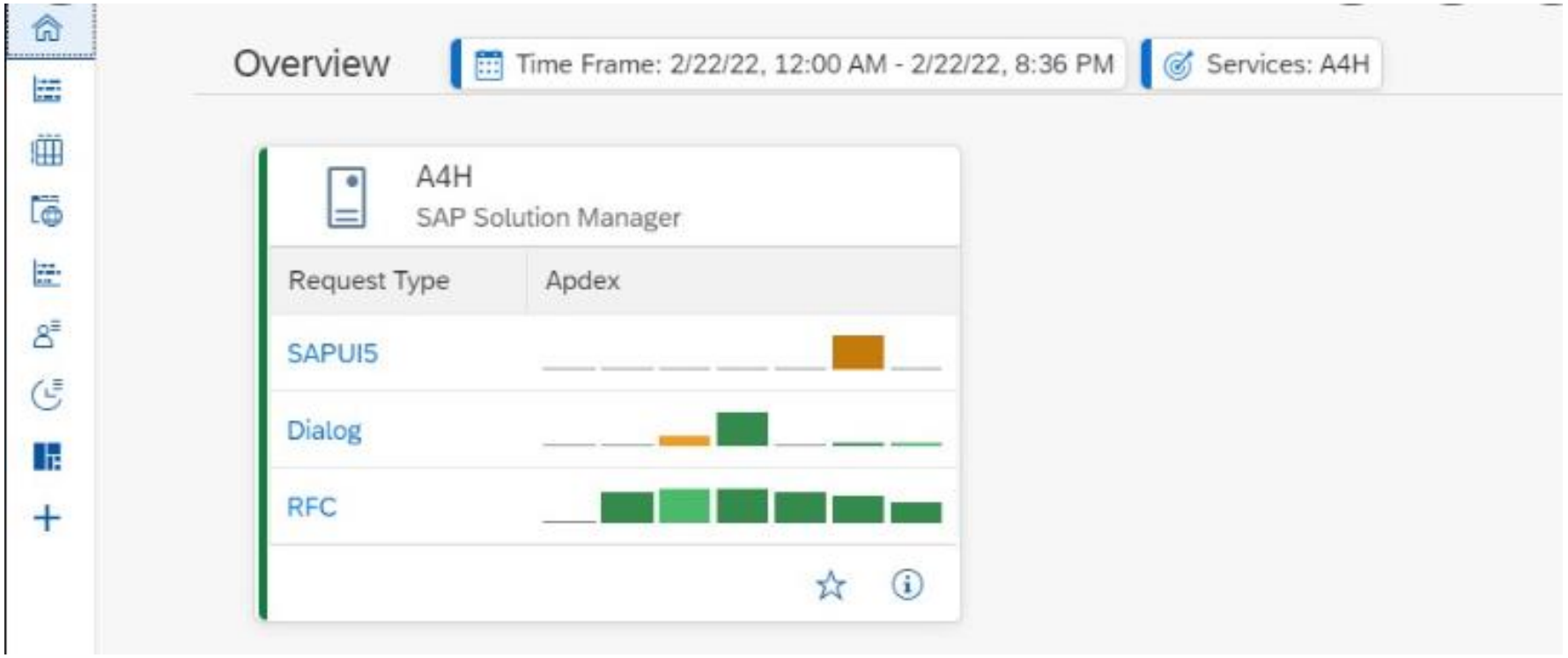
SAP Cloud ALM for operations

Real User Monitoring

Real User Monitoring



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SAP Cloud ALM for operations

Health monitoring

Overview | Favorites

Favorites

SAP Cloud ALM
Services: 1
Ok 1
Quality | Alerts 0

SAP Integration Su...
Services: 2
Ok 2
Quality | Alerts 0

Health Monitoring

Monitoring | Scope: cloudalm, production-kuh376fl

Home / Monitoring / production-kuh376fl (SAP Integration Suite (Cloud Integration)) /

production-kuh376fl (SAP Integration Suite (Cloud Integration))

Description: production-kuh376fl (0M4EO4J)

Root URL: [redacted]hana.ondemand.com/itspa

Tenant Type: Production | Customer Name: 12485

Tenant ID: 742875283 | Customer Number: 12485

Metrics Overview

Certificates

Certificate Validity
Ok 8

JMS Queues

No data

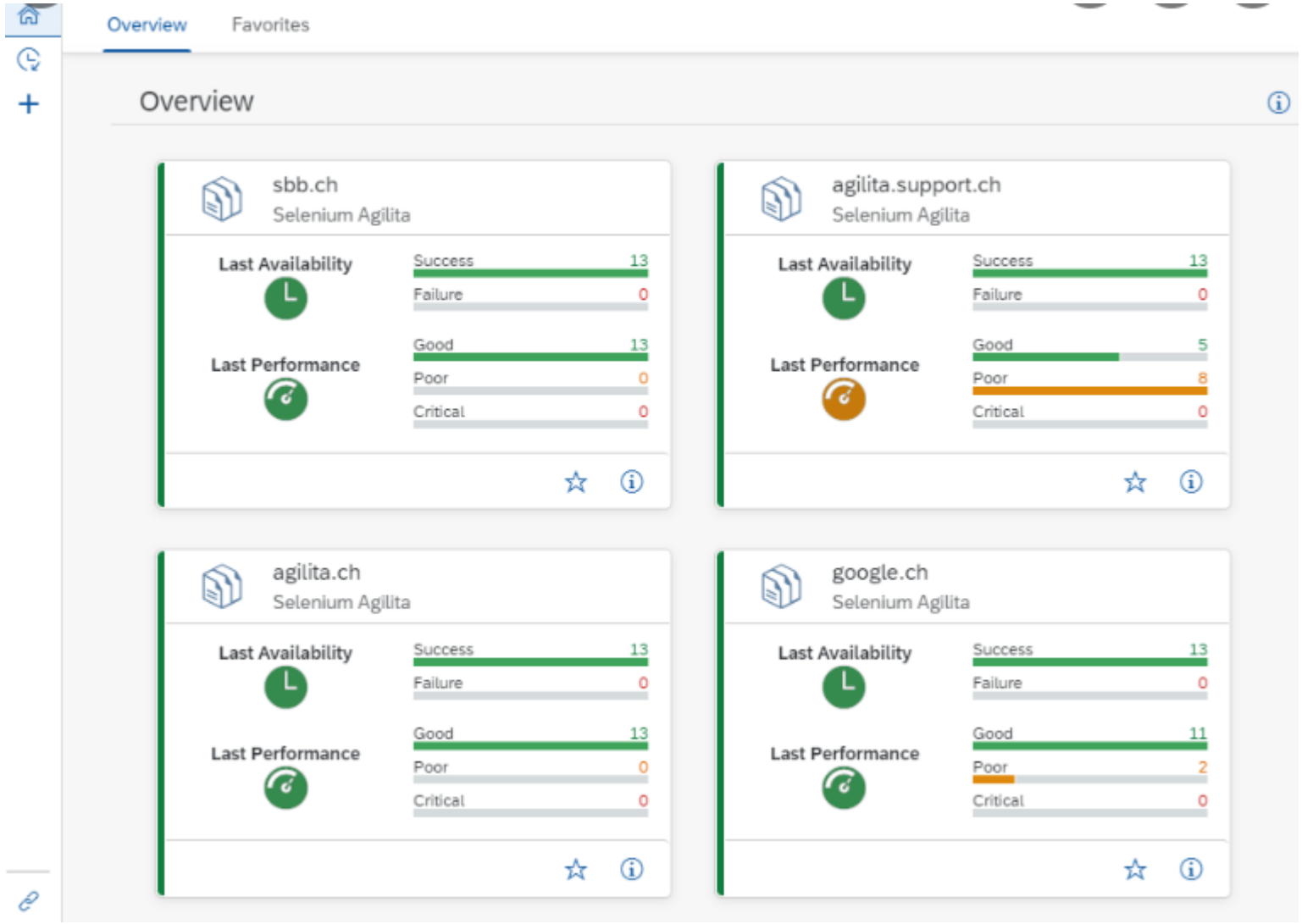
Search to filter metric name, label or text

Name	Alias	Type	Rating	Value	History
Certificate Validity	sap_cloudintegrationcertificate	KeyPair		340 days	
Certificate Validity	cpi_sfapi_oauth	KeyPair		728 days	
Certificate Validity	sap_baltimorecybertrustroot	Certificate		961 days	
Certificate Validity	sap_digicertglobalcag2	Certificate		2138 days	
Certificate Validity	sap_digicertglobalrootca	Certificate		3333 days	
Certificate Validity	sap_verisignclass3publicprimarycertificationauthority-g5	Certificate		5044 days	
Certificate Validity	sap_verisignuniversalrootcertificationauthority	Certificate		5547 days	
Certificate Validity	sap_digicertglobalrootg2	Certificate		5592 days	

SAP Cloud ALM for operations

Synthetic user monitoring

Synthetic User Monitoring



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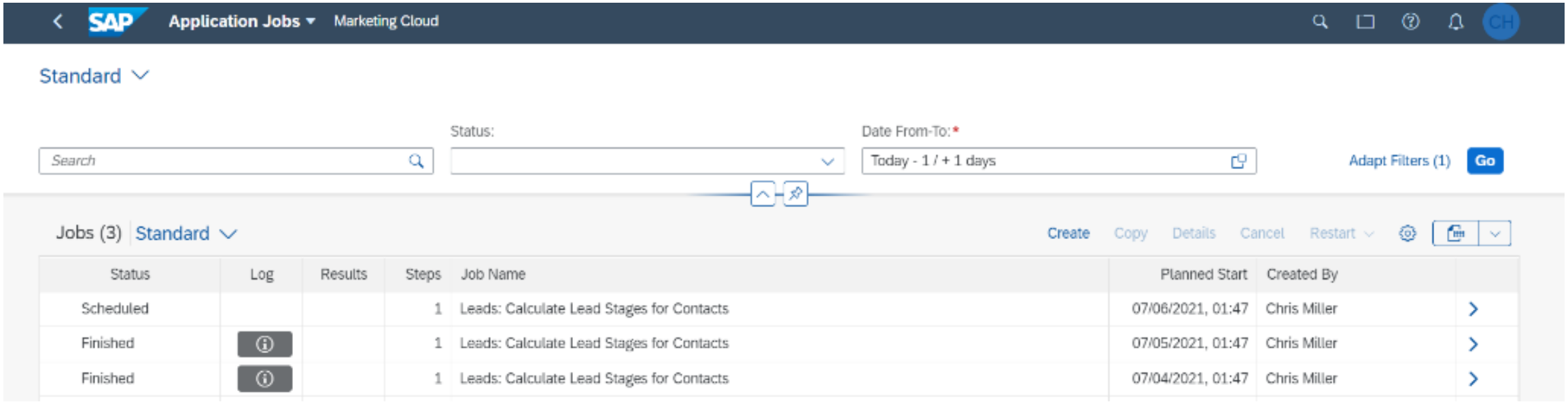
SAP Cloud ALM for operations

Job Automation and Monitoring

Job & Automation Monitoring



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SAP Cloud ALM for operations

Intelligent event processing

Intelligent Event Processing

Event Processing Rules

Rules (18)

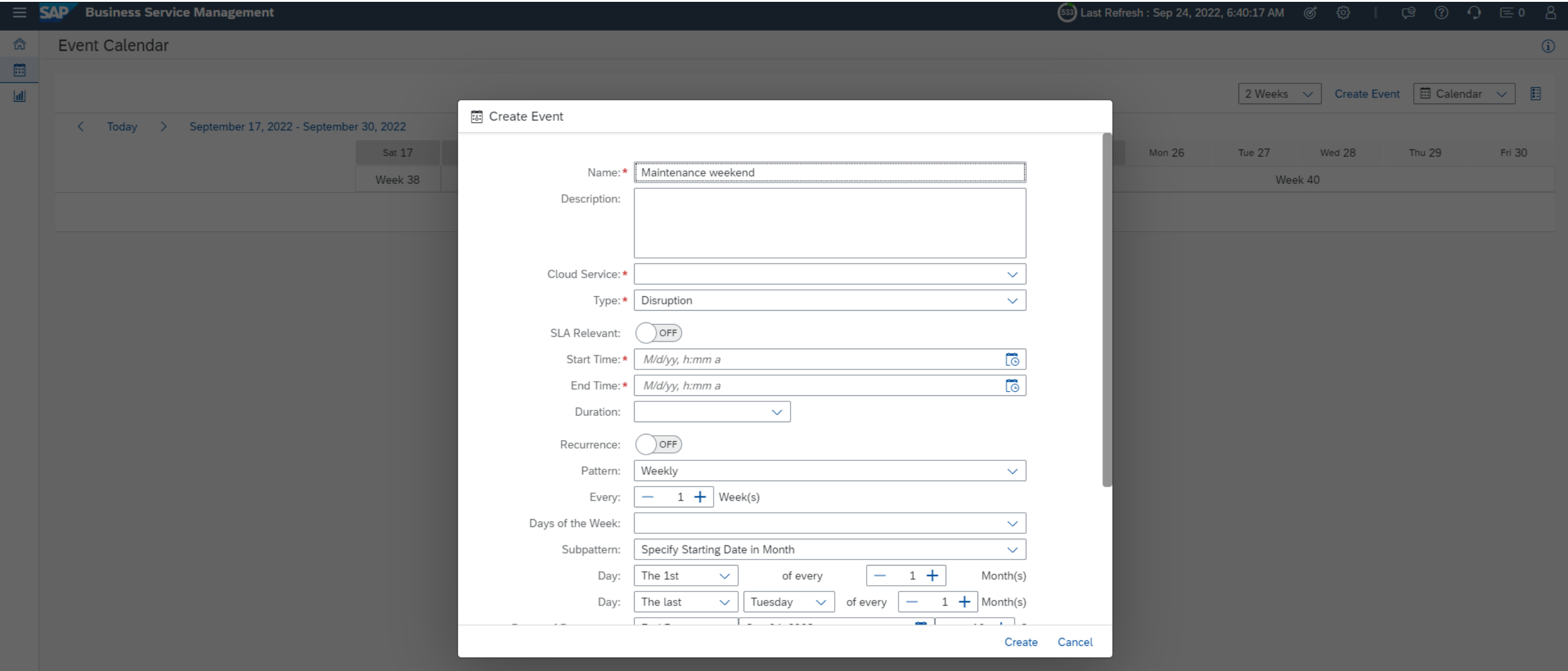
Search Rules

Name	Status	Origin	Last Changed By	Last Changed At
CALM.IM_calm.im.cpi.msgs	✓	Integration Monitoring	[redacted]	Sep 18, 2022 12:33:08 PM
CALM.IM_calm.im.cpi.msg	✓	Integration Monitoring	[redacted]	Sep 07, 2022 11:39:04 AM
CALM.IM_calm.im.cpi.msg	✓	Integration Monitoring	[redacted]	Sep 07, 2022 11:39:04 AM
CALM.IM_calm.im.cpi.msgs	✓	Integration Monitoring	[redacted]	Sep 07, 2022 11:21:47 AM
CALM.IM_calm.im.cpi.msgs	✓	Integration Monitoring	[redacted]	Sep 07, 2022 11:21:47 AM
CALM.EXM_STATEFUL_CPI	✓	Integration Monitoring	[redacted]	Sep 07, 2022 11:17:32 AM
CALM.HM_CPI.Certificates.Expired	✓	Health Monitoring	[redacted]	Aug 08, 2022 11:54:29 AM
CALM.HM_CPI.Certificates.Expired	✓	Health Monitoring	[redacted]	Aug 08, 2022 11:54:29 AM
CALM.IM_calm.im.sf.msgs	✓	Integration Monitoring	[redacted]	Jul 20, 2022 02:00:35 PM
CALM.IM_calm.im.cpi.msg	✓	Integration Monitoring	[redacted]	Jul 20, 2022 11:54:21 AM
CALM.IM_calm.im.cpi.msg	✓	Integration Monitoring	[redacted]	Jul 20, 2022 11:54:21 AM
CALM.EXM_STATEFUL_CPI	✓	Integration Monitoring	[redacted]	Jul 20, 2022 11:52:10 AM
CALM.IM_calm.im.cpi.msgs	✓	Integration Monitoring	[redacted]	Jul 19, 2022 04:47:16 PM
CALM.IM_calm.im.cpi.msgs	✓	Integration Monitoring	[redacted]	Jul 19, 2022 04:47:16 PM
CALM.IM_calm.im.cpi.msg	✓	Integration Monitoring	[redacted]	Jul 19, 2022 04:46:23 PM
CALM.IM_calm.im.cpi.msg	✓	Integration Monitoring	[redacted]	Jul 19, 2022 04:46:23 PM
CALM.EXM_STATEFUL_CPI	✓	Integration Monitoring	[redacted]	Jul 19, 2022 02:59:37 PM
CALM.EXM_STATEFUL_CPI	✓	Integration Monitoring	[redacted]	Jul 19, 2022 02:59:37 PM

SAP Cloud ALM for operations

Business service management

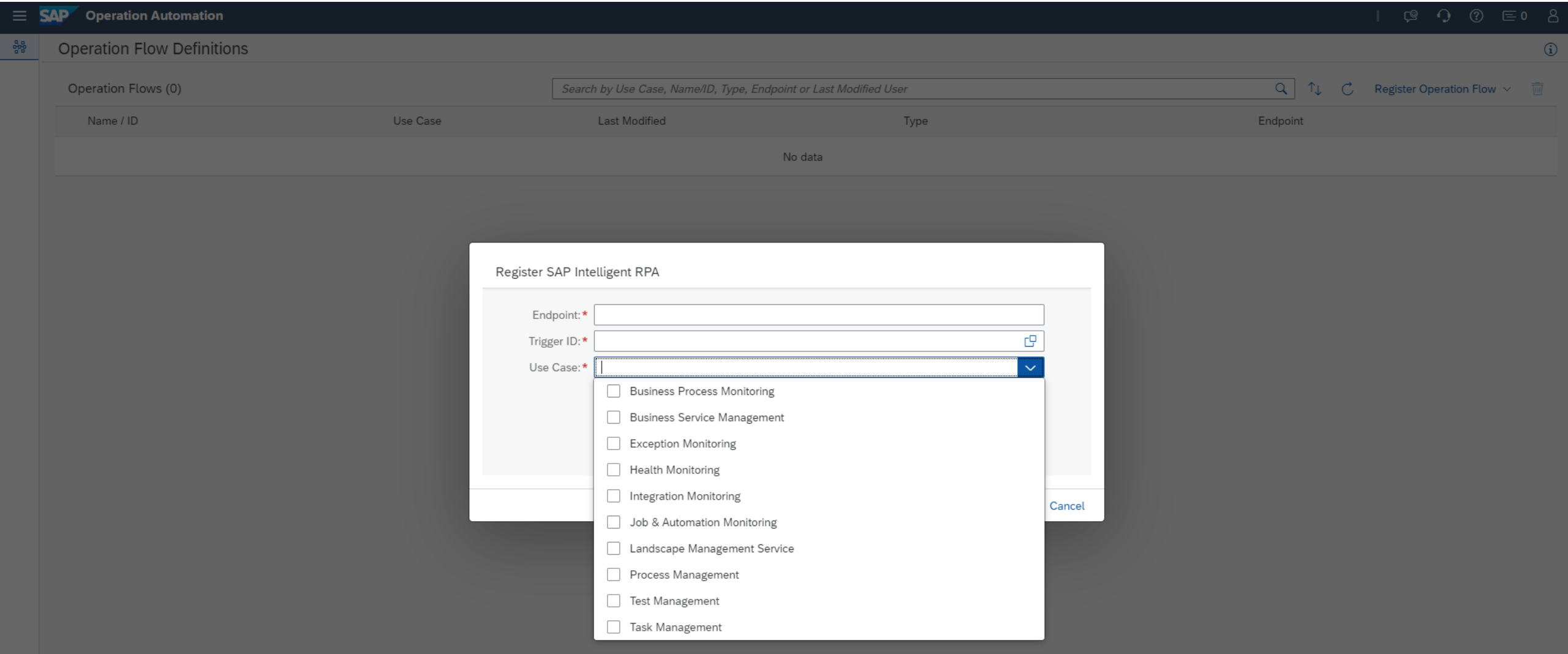
Business Service Management



SAP Cloud ALM for operations

Operation automation

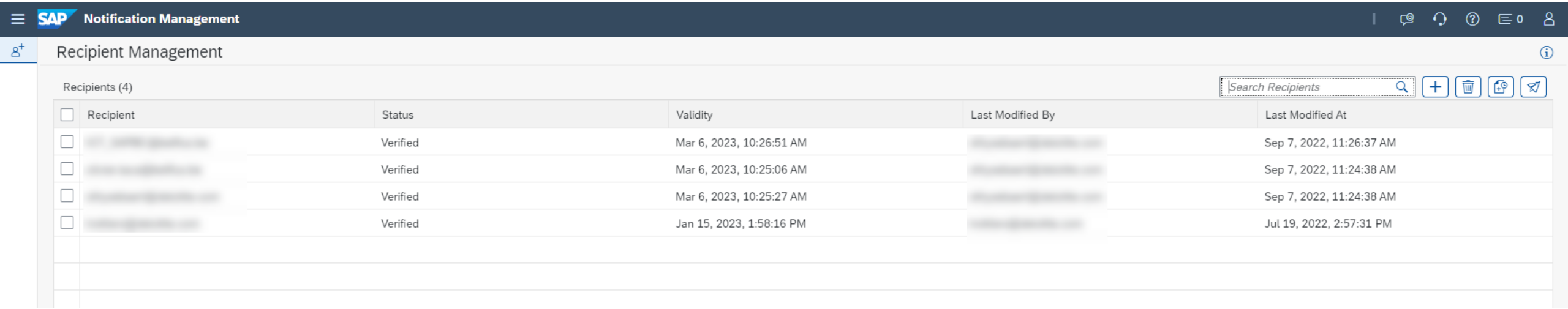
Operation Automation



SAP Cloud ALM for operations

Notification management

Notification Management



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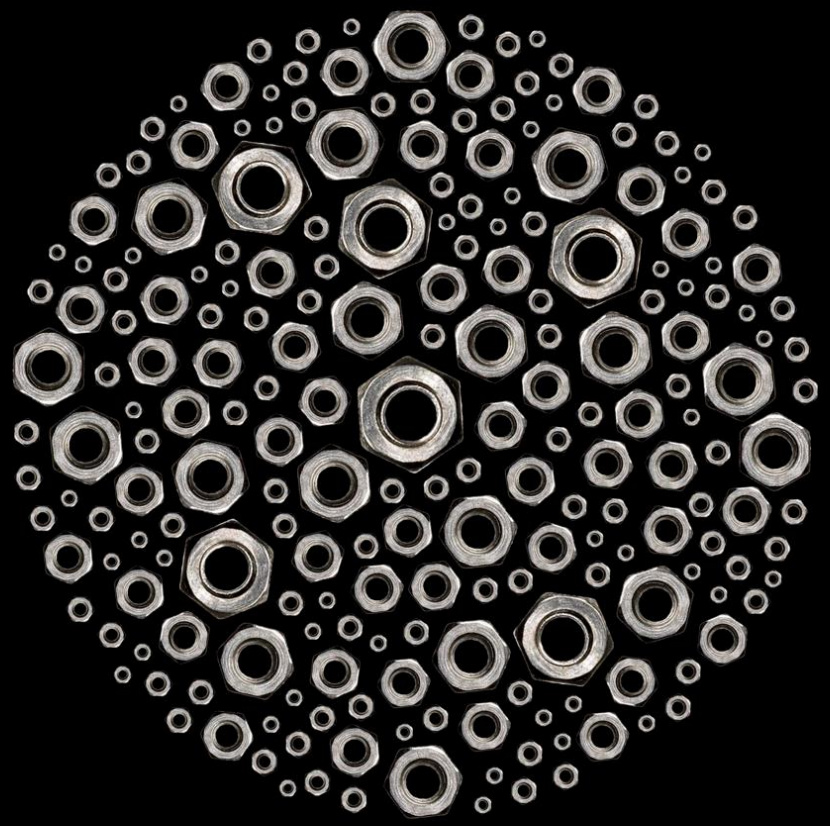

SAP Cloud ALM API’s

SAP Cloud ALM API's

Starting point

https://blogs.sap.com/2021/08/05/sap-cloud-alm-extend-with-apis-introduction-use-cases/

Use cases

Consume data in 3rd party reporting tools/custom applications

Integration with 3rd party tools:

- Test management
- Project/task management
- CI/CD
- Collaboration

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SAP Cloud ALM API’s

API business hub

https://api.sap.com/package/SAPCloudALM/rest

SAP API Business Hub | Explore | Resources | Discover Integrations | Partner with Us | Login

REST API

SAP Cloud ALM Analytics

The SAP Cloud ALM Analytics API is used to unify access to all analytics data produced by SAP Cloud ALM.

Version 1.0.0 ACTIVE

REST API

SAP Cloud ALM Business Service Management

Get downtime events (disruptions, degradations, and maintenance) from Business Service Management

Version Beta BETA

REST API

SAP Cloud ALM Event Situations

Allows to close, update event situations

Version Beta BETA

REST API

SAP Cloud ALM Projects

Allows to get or create projects in project management.

Version 1.0.0 ACTIVE

REST API

SAP Cloud ALM Resource Changes

The SAP Cloud ALM Resource Changes API notifies external third party applications on SAP Cloud ALM resource changes.

Version 1.0.0 BETA

REST API

SAP Cloud ALM Tasks

Allows to get, create, modify or delete tasks in task management.

Version 1.0.1 ACTIVE

REST API

SAP Cloud ALM Test Automation

Allows cloud-based test automation providers to integrate with SAP Cloud ALM.

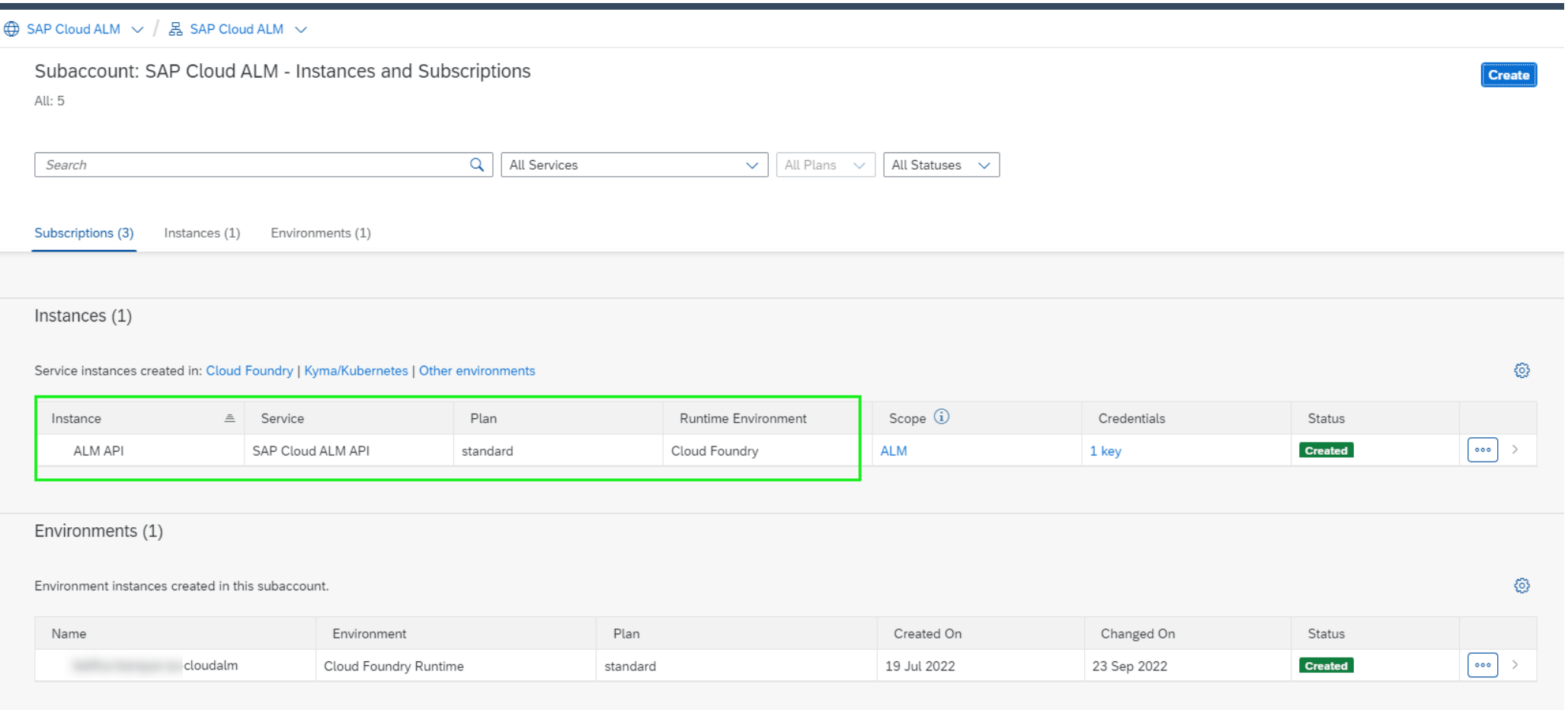
Version 1.0.10 ACTIVE

FEEDBACK

SAP Cloud ALM API’s

Enabling the API’s

Create SAP Cloud ALM API service instance and assign relevant scopes:



SAP Cloud ALM API’s

Enabling the API’s

Create SAP Cloud ALM API service instance and assign relevant scopes:

```
{
 "xs-security": {
  "xsappname": "4a359440-1748-49df-999a-0267a2dbc699",
  "authorities": [
   "$XSMASTERAPPNAME.calm-api.analytics.read",
   "$XSMASTERAPPNAME.calm-api.analytics.providers.read",
   "$XSMASTERAPPNAME.calm-api.demo-tasks.read",
   "$XSMASTERAPPNAME.calm-api.projects.read",
   "$XSMASTERAPPNAME.calm-api.tasks.read",
   "$XSMASTERAPPNAME.calm-api.projects.write",
   "$XSMASTERAPPNAME.calm-api.tasks.write"
  ]
 }
}
```

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SAP Cloud ALM API’s

Enabling the API’s

SAP Cloud ALM / SAP Cloud ALM

Subaccount: SAP Cloud ALM - Instances and Subscriptions

Create

All: 5

Subscriptions (3)

Cloud Integration Automati...	Subscribed	
SAP Cloud ALM	Subscribed	

Instances (1)

Service instances created in: Cloud Foundry | Kyma/Kubernetes | Other environments

Instance	Status	
ALM API	Created	

Environments (1)

Environment instances created in this subaccount.

Name	Status	
belfius-banque-sa-cloudalm	Created	

ALM API

View Credentials

Instance ID: **4a359440-1748-49df-999a-0267a2dbc699**
Service: SAP Cloud ALM API (SAPCloudALMAPIs)
Plan: **standard**

Runtime Environment: **Cloud Foundry**
Space: ALM
Status: Created

Created On: **23 Sep 2022**
Changed On: **23 Sep 2022**

Bound Applications (0) | Service Keys (1) | Labels (0)

Create

Name	Status
No bound applications.	

Service Keys (1)

Create

Name	Status	
ALM API	Created	

Labels (0)

Add

No labels assigned to this instance. Click 'Add' to assign labels.

SAP Cloud ALM API’s

Enabling the API’s

Cloud ALM / **ALM analytics providers**

GET [redacted]/api/calm-analytics/v1/analytics/providers

Params | Authorization ● | Headers (7) | Body | Pre-request Script | Tests | Settings

Type OAuth 2.0

Current Token

This access token is only available to you. Sync the token to let collaborators on this request use it.

Body | Cookies | Headers (8) | Test Results

Pretty | Raw | Preview | Visualize | JSON


```
[
    {
        "name": "PROVIDERS_LIST",
        "description": "ALM Analytics Providers",
        "service": "x-calmextapi-srv",
        "domain": "Analytics",
        "scope": "calm-api.analytics.providers.read",
        "version": [
            "v1"
        ],
        "plan": "standard",
        "calmMarket": "RUN"
    },
    {
        "name": "DEMO_TASKS",
        "description": "ALM Demo Tasks",
        "service": "x-calmextapi-srv",
```


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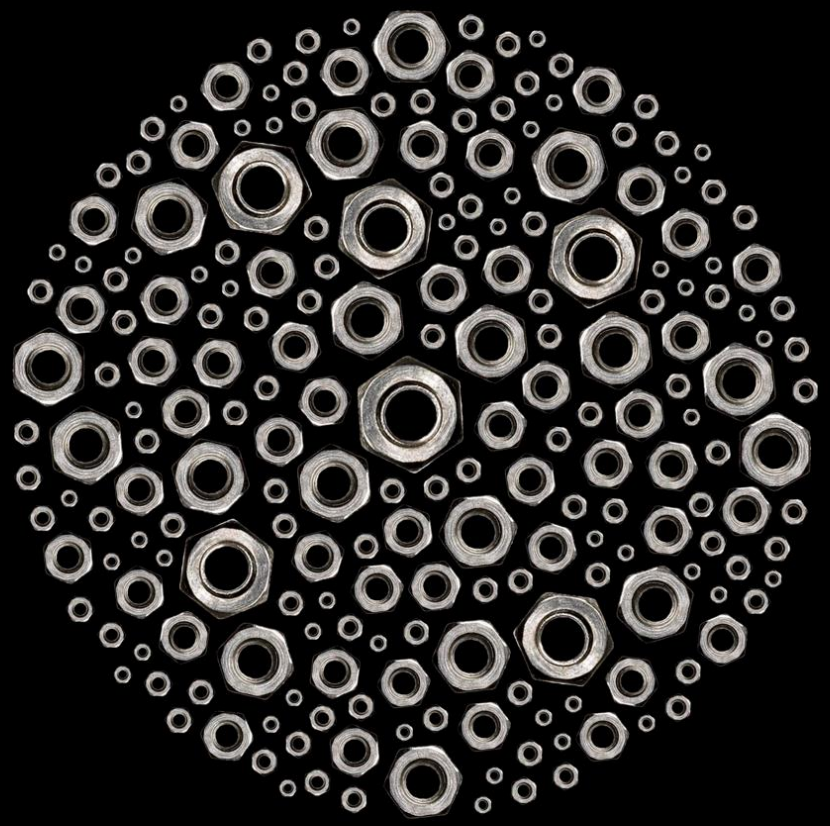

Conclusion

Conclusion

- Free
- Easy to implement (2 hrs)
- API’s for extensibility
- Many out of the box capabilities

- Limited in functionality/options

What’s next for me?

- Further exploring of the API’s
- ALM for implementation

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